Horizon — how the whole pipeline works

green = math · lavender = AI model · rose = guard / attack tool · white = program code · beige = data

① BEFORE — prepared once (offline)

Datasets
indirect: BIPIA · LLMail-Inject
direct: HackAPrompt · HarmBench · JBB · Gandalf · XSTest · Alpaca

Stage 1 experiment
every attack document × 8 markings τ
(delimiters · ^ · N^k · ROT13 · shift · scramble · Base64)
new secret key κ each time · Qwen3-8B answers

Stage 1 maths
attack rate bound $\bar{p} = \text{Beta}^{-1}(1-\alpha; s+1, n-s)$
χ^2-DRO $g_\rho(\bar{p}) = \bar{p} + \sqrt{(2\rho\bar{p}(1-\bar{p}))}$
forging $q/|K_\tau|$ · adaptive attacker $\bar{p}^{adp}$
$R_\tau = \min\{1, q/|K_\tau| + \max(g_\rho, \bar{p}^{adp})\}$

Sion game (level 1)
$V_1 = \min_\pi \max_{attack}$ (cvxpy SOCP)
→ **π*** = best random mix of markings

Calibration
conformal risk control on agent runs
$\hat{\lambda}$ = smallest λ with $(n/(n+1))\hat{R}(\lambda) + 1/(n+1) \le \alpha_{user}$

Rule table proof
Z3 proves config/policy.yaml:
untrusted data ⇒ never ALLOW

π* (which markings, how often)

$\hat{\lambda}$

proven rules

② DURING — one request, step by step (online)

User request
e.g. “summarise my inbox” (trusted)

1 · Input guard
DeBERTa injection classifier
blocks a jailbreak / injection typed by the user

2 · Planner AI (P-LLM)
sees ONLY the request, never outside data
writes a small program

3 · CaMeL interpreter
runs the program, calls tools,
remembers where every value came from

reads untrusted data

Outside data
emails · web pages · tool results
(UNTRUSTED — may hide an attack)

4 · Reader AIs (Q-LLM) × m
each reader: own marking τ ~ π* and own key κ
no tools · must answer in a fixed format

Leakage
≤ Σ $\log_2|S|$ bits per answer
≤ leakage_cap_bits

5 · Majority vote
> m/2 readers agree → value, else refused

Hoeffding
P(vote fooled) ≤ $\exp(-2m(½ - \bar{R}_Q)^2)$

6 · Label the value
SYSTEM ⊑ USER ⊑ UNTRUSTED
mixed data keeps the strictest label

Label join
$\text{cap}(f(v_{1..k})) = \sqcup \text{cap}(v_i)$

7 · Rule check
rule table per tool

decision?

CONFIRM

DENY

ALLOW

8 · Ask or deny?
readers' agreement $s(x) \ge \hat{\lambda}$?

yes

no

ask YOU
approve / decline

DENY
action blocked

approved

9 · Tool runs
AgentDojo world (simulated, safe)

10 · Output guard
Qwen3Guard (harmful?) + canary
(secret code word leaked?)

Answer

③ AFTER — test and measure

Attack tools
AgentDojo injections · PyRIT XPIA (indirect)
PyRIT PAIR · TAP · Crescendo (jailbreaks)
garak probes · guard detection on datasets

5 defence setups (same attacks)
none · stage1 · stage2 · stage1+2 · full
each test = one request through ②

Count & bound
attack got through? → ASR = s/n
95 % Clopper–Pearson interval · McNemar · bootstrap
loss $\ell = G + a{\cdot}C + h{\cdot}F$

Sion game (level 2) + Horizon value
min over π∈Π, max over w of Σ w ℓ (scipy LP)
$V_H = \min\{1, \Sigma\pi\, q/|K| + \max(E[L] + \sqrt{(2\rho\ \text{Var}\ L)}, \Sigma\pi\, \ell^{adp})\}$

Results
./run_campaign.sh → report/campaign/ (process + through / blocked)
figures F1–F18 · report/report.pdf (written by the colibri model) · MLflow